DOCTOR OF BUSINESS ADMINISTRATION | RESEARCH PROPOSAL

GIFT RE

An AI-Powered Global Reinsurer

Synthetic data for risk pricing. ML for reinsurance asset management. Built at GIFT City. Catering to the needs of emerging economies

Presented by **Abhay Tewari**
Doctor of Business Administration in Emerging Technologies
Golden Gate University · Cohort 6894
Chair: Dr. Siwar Jendoubi · May 2026

PRESENTER



Abhay Tewari

MD & CEO · Star Union Dai-ichi Life Insurance

FIA (UK) · FIAI · CFA

ABOUT THE RESEARCHER

28+ years at the intersection of Actuarial Science, Investment and AI

CAREER

MD & CEO, SUD Life
2021 – present · Revenue CAGR >25%

Chief Actuary, SUD Life
2014– 2021

Appointed Actuary, Edelweiss Life
2010 – 2014

Dy. Chief Actuary, SBI Life
Earlier career · SEBI, Surveillance & Investigation

EDUCATION

PGP AIML – University of Texas
AIML Models and application

IIT Kanpur
E-Masters in Economics, Finance and Public Policy

FIA · Institute & Faculty of Actuaries (UK)

FIAI · Institute of Actuaries of India

CFA Charter · CFA Institute (USA)

RESEARCH & VENTURES

Gift Re
AI-enabled reinsurer · GIFT City

GanitagyAI
Actuarial software · IFRS 17 / IndAS 117

AIF Platform
Cat 1 & Cat 3 funds · InsurTech / AI

Agenda

01

Context & Opportunity

The reinsurance opportunity and the case for Gift Re

02

Research & Industry Insights

Evidence from industry and academic research, customer insights— converging on actionable opportunities for Gift Re

03

Solutions Propositions & Evidence

Two solution propositions — ML-driven alpha and improved CAT pricing — validated through expert interviews and back-testing

04

Competition, Financials & Roadmap

Competitive Landscape, Financials, Technology Architecture & MVP Roadmap

05

Team, Capital & SWOT

Team structure, Risk Based Capital, SWOT analysis

06

Business Impact, Risks & Key Takeaways

Business Impact & Research Contribution, Risks & Mitigation, Key Takeaways

What is reinsurance — the risk-transfer chain

INSURER
Primary Carrier

Issues policies to individuals and businesses. Retains manageable risk, cedes excess to reinsurer.

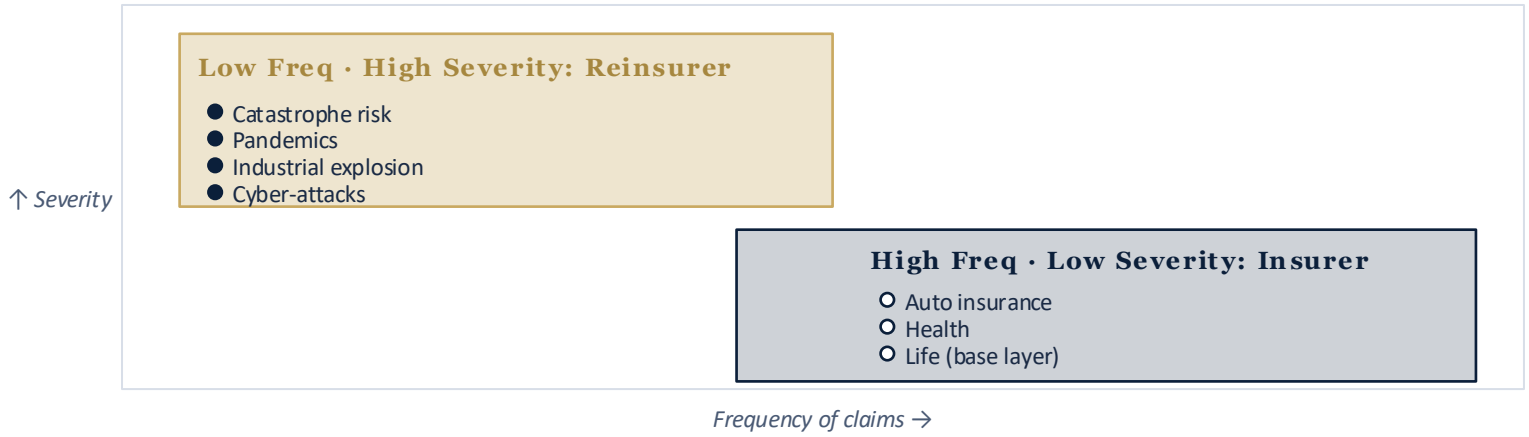
REINSURER
Secondary Risk Taker

Assumes a portion of risk from insurers. Spreads risk across geographies and lines. May cede further.

RETROCESSIONAIRE
Reinsurer's Reinsurer

Provides coverage to reinsurers. Caps reinsurer exposure to extreme, correlated loss events.

Risk appetite · who takes what



Two unresolved problems on a reinsurer's balance sheet

THE LIABILITY SIDE

Pricing what you can't model

- Catastrophe risks are rare, severe, and thinly documented — especially in emerging markets
- CAT models calibrated on US/Atlantic data fail on Indian flood, cyclone, and monsoon risks
- Actuaries add large adverse-deviation margins → premiums become unaffordable
- The protection gap in tier-3/4 India keeps widening

THE ASSET SIDE

Generating investment alpha

- Reinsurance float is ~2x capital — large, persistent, investable capital
- Most reinsurers invest conservatively: short-tenor government bonds, zero exposure to equities. Berkshire is an exception, with high equity exposure
- ALM runs in silos — quarterly, rule-based, not integrated with claims distribution
- Hedge funds have been using ML for alpha; reinsurers have not

Vision, Mission & USP

Gift Re — An AI-Powered Global Reinsurer at GIFT City, India

Synthetic data for risk pricing · ML for reinsurance float · Integrated ALM · Built for emerging markets

VISION

To lead the reinsurance market in developing countries, transforming risk and investment management through technology — handling complex catastrophic risks while maximising returns on capital.

MISSION

To enable insurers with technology-driven reinsurance solutions, enhancing risk resilience and financial security — aligned with India's "Insurance for All by 2047" and the needs of underinsured emerging economies.

UNIQUE SELLING PROPOSITIONS

1 Synthetic data alongside historical claims to price catastrophe risks where data is scarce

2 ML-driven investment management to generate superior risk-adjusted returns on reinsurance float

3 Fully integrated ALM platform where claims distribution, liquidity needs, and solvency capital feed into asset duration and mix across money market, debt, and equity

4 Proprietary models purpose-built for tier-3/4 India and other underinsured emerging economies

Product portfolio — covering the full reinsurance lifecycle



Life & Health

Mortality, morbidity, longevity, critical illness. Target: LIC, private life insurers and general insurers. Stand alone health insurers (SAHI)



Property & Casualty

Fire, earthquake, flood, cyclone. Parametric triggers built in for tier-3/4 rural portfolios.



Treaty

Proportional and non-proportional for portfolio-level risk transfer. Multi-year renewals for stable relationships.



Facultative

Case-by-case coverage for specific large or unusual risks falling outside treaty terms.



Parametric

Trigger-based payouts — wind speed, rainfall, magnitude. Instant settlement for catastrophe response.



Crop & Cyber

AI-risk liability, data-breach, ransomware. Priced via synthetic scenarios where historical data is absent.

Why now - increasing requirements, emerging solutions

Climate losses rising

Insured nat-cat losses reached USD 137B in 2024 and are projected to cross USD 145B in 2025. Frequency and severity, both are accelerating.

Improved Generative AI applications

GANs, VAEs, and Temporal Fusion Transformers can now generate reliable synthetic tail scenarios and explain their outputs to actuaries and regulators.

India is fast but significantly underinsured

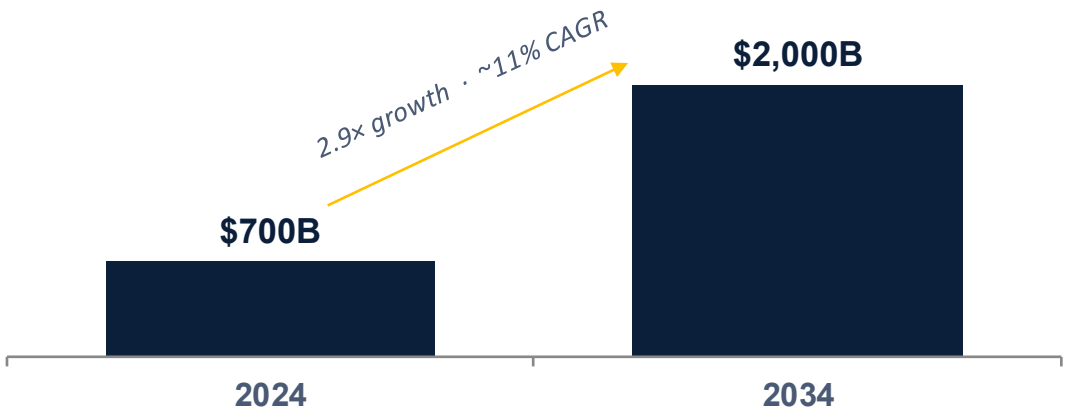
Indian insurance revenues have grown >10% CAGR for 20+ years even though rates have fallen by more than 10%. Tier-3 / tier-4 markets remain underserved; foreign reinsurers stay away.

Regulatory tailwinds in India and specially at GIFT City

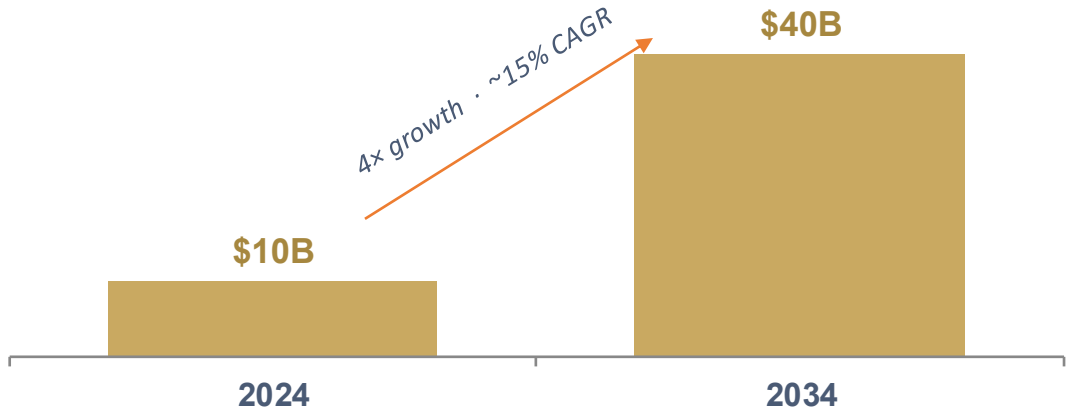
100% FDI, Goods and Service Tax (GST) removed. IFSCA framework enables cross-border reinsurance; India is moving from rule-based to Risk-Based Capital (RBC) — rewarding better modelling. Public Insurance Registry envisaged to reach underserved segments

The protection gap — 18% people, less than 2% of reinsurance

GLOBAL REINSURANCE



INDIA REINSURANCE



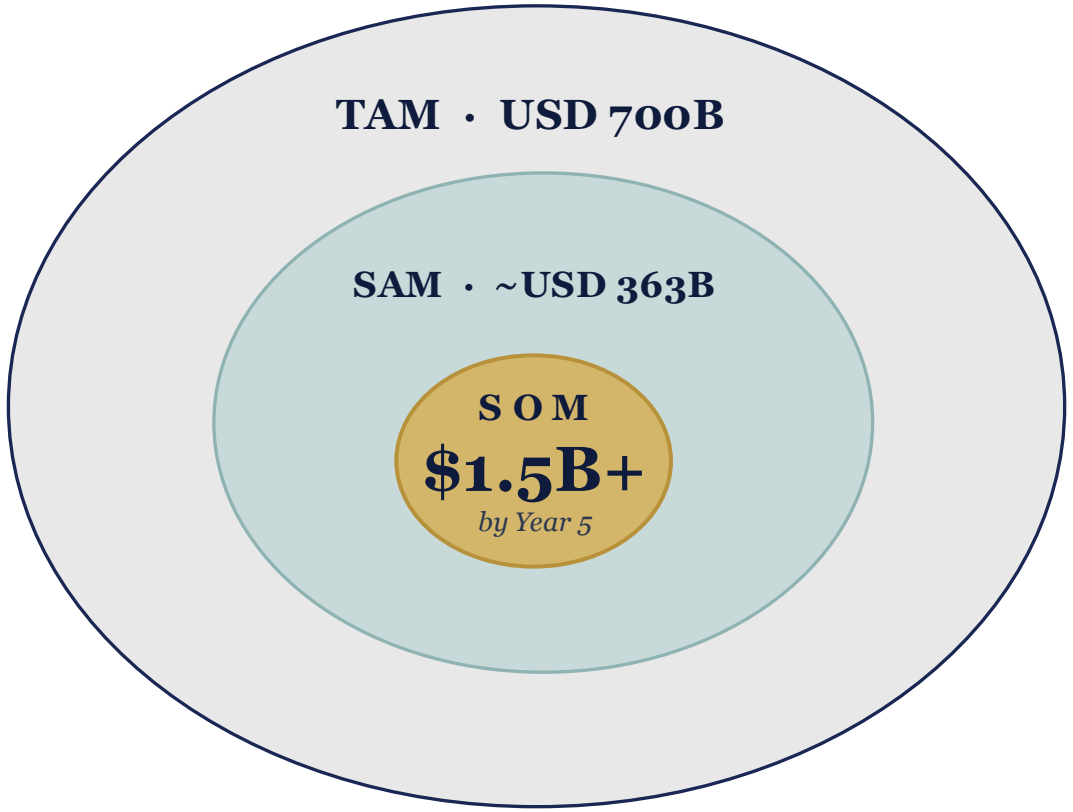
THE PROTECTION GAP

~18%
India's share of global population

1.4% → 2%
India's share of global reinsurance (2024 → 2034)

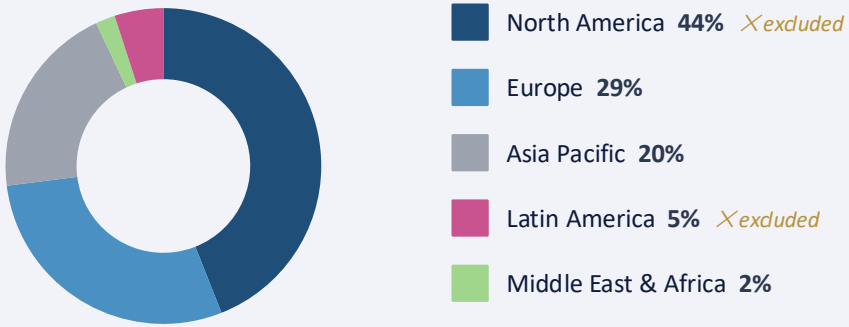
Sources: Precedence Research (2025); UN World Population Prospects (2024)

TAM · SAM · SOM — a credible path to USD 4B+ by year 10



Global reinsurance market by region · 2024

North America + Latin America = 49% → excluded from SAM



Source: Precedence Research (2025)

SOM · Year 5

~USD 1.5B

$1.0B Asia-Pacific + $0.5B other regions, as digital, parametric, and embedded placements scale.

SOM · Year 10

$4B+

$2.0B Asia-Pacific + $2.0B other regions, assuming Indian market continues at ~15% CAGR and GIFT City gains global standing.

01 · CORE PROPOSITION

AI to make reinsurance more effective

Specifically: *Whether ML–driven investment management can generate superior risk-adjusted returns, and whether generative models (GANs/VAEs) can meaningfully improve the catastrophe risk pricing in data-scarce markets.*

SP1

ML for float management

Asset side · return generation

SP2

Synthetic data for risk pricing

Liability side · better pricing

PART TWO

Research & Industry Insights

Evidence from industry and academic research, customer and practitioner's insights—converging on actionable opportunities for Gift Re

What the Evidence Says — Two Streams, One Opportunity

Liabilities stream - Generating synthetic data to improve risk pricing

- Hurricane Andrew (1992) revealed that pricing each peril in isolation with conservative margins fails when multiple hazards interact simultaneously — giving rise to the structured Hazard · Vulnerability · Exposure · Financial Loss framework.
  BIS (2025); Golnaraghi et al. (2018)
- Traditional Generalised Linear Models (GLMs) capture individual risk factors but cannot model non-linear interactions or joint tail behaviour across correlated perils — leaving multi-peril portfolios priced inaccurately.
  Burnecki et al. (2024)
- Wasserstein GANs (WGAN-GP) preserve heavy-tailed distributions, enabling reliable synthesis of joint extreme-loss scenarios for reinsurance pricing.
  Gu et al. (2025)
- In emerging markets, thin claims history limits traditional calibration. Satellite imagery, geospatial demographics, and real-time environmental data can augment scarce historical records to build credible catastrophe risk models for tier-3/4 India and similar markets.
  Guin et al. (2024)

Assets Stream · Better float management

- Amongst Hedge-funds, ML usage is quite high, (re)insurance usage seems to absent or negligible
  (Peterson et al., 2025)
- Most reinsurers are focused on liquidity and invest in short-duration bonds, with no equity exposure
  (AI Harrison et al., 2025)
- Berkshire earns alpha from fundamentals; no ML applied to float
  (Buffett et al., 2024)
- Machine learning forecasts can double the performance of traditional investment strategies, generating significant economic gains and higher Sharpe ratios. BiLSTM networks excel at capturing complex forward and backward temporal dependencies in highly volatile financial markets
  (Gu et al., 2020); (Zhang et al., 2024)

Customer Discovery — Six Themes from 20+ Industry Leaders

01

Investment management is an under-exploited lever

Multiple CEOs and Appointed Actuaries said better returns on float — even a few basis points — compound into a meaningful competitive edge over time.

02

Traditional ALM is broken in two ways

Nominal-bond portfolios create real-return mismatches on inflation-sensitive claims, and short-tail duration matching forces reinsurers into sub-optimal low-yield T-bills.

03

ML can beat linear models — but explainability is a hard constraint

Actuaries will reject any model they cannot explain to IRDAI. Random Forests and TFTs are accepted more readily than deep neural nets.

04

Catastrophe models break under emerging-market conditions

A 2% error in vulnerability assumptions can swing PML by 50%. CAT models calibrated on US data consistently misprice Indian flood and cyclone risk.

05

Synthetic data is promising — but can be a complement, not a substitute

GANs work on known-pattern rare events. For truly novel risks (e.g., AI-driven cyberattack) they can produce realistic-looking but meaningless output.

06

AI-enabled reinsurer will have a structural advantage

Building infrastructure from scratch enables real-time ALM, daily ML-driven rebalancing, and alternative-data underwriting for undocumented populations.

The White Space Gift Re is Built to Fill

01 Synthetic-data pricing is not operationalised

Reinsurance companies are using AI-ML to facilitate the underwriting, but synthetic data creation has not been applied. Integration of Extreme Value Theory and Generalized Linear Models with the technologies is not well covered in the existing literature.

02 ML for float management is a hedge-fund, not a reinsurer

Deep learning and reinforcement learning are extensively used by hedge funds, however in reinsurance it is limited, especially the approach to optimize the returns while balancing reinsurance-specific constraints like liability duration, solvency capital and liquidity still remain unexplored.

03 Emerging-market catastrophe modeling is incomplete and not transferable

Existing CAT models are largely calibrated on US/Atlantic hazard, vulnerability, and exposure data. The underlying assumptions do not transfer to Indian flood, cyclone, or monsoon conditions — making them incomplete for emerging markets and not transferable across geographies.

04 Explainability gap

Reinsurance is heavily regulated — every pricing assumption must be certifiable by an Appointed Actuary. Deep learning models generate outputs without auditable causal chains. No incumbent has operationalised a hybrid ML → GLM pipeline that satisfies both predictive accuracy and regulatory explainability

PART THREE

Solution Propositions & Evidence

Two solution propositions — ML-driven investment alpha and improved CAT pricing — validated through 20+ expert interviews and backtesting of the investment MVP.

Solution Proposition 1 — ML for better alpha and superior risk-adjusted returns on float

SP1

Machine-learning–based investment management of reinsurance float *— using models such as BiLSTM, CNNs, Random Forests, and Temporal Fusion Transformers — are applied and compared with traditional benchmark indices* ***to see if ML models can generate superior risk-adjusted returns****.*

Constructs

- Reinsurance float (pooled investable assets)
- Risk-adjusted return (Sharpe, Treynor)
- Solvency & liquidity constraints
- Business-cycle regime

Methods

- BiLSTM, CNN, Random Forest, Logistic regression
- 16+4 years (Training + Testing) of data on top 1,000 NSE companies on Market Cap
- Quarterly rebalance portfolio
- Portfolio optimized via Sharpe

Success metrics

- CAGR vs Nifty 200 factor indices
- Sharpe / Treynor uplift
- Jensen's Alpha and Information Ratio
- ROCE improvement under RBC

16 Quarters back test — ML beats every factor index

Strategy	CAGR	StdDev	Beta	TE	Sharpe	Treynor	Alpha	IR
Nifty 50 (benchmark)	12.2%	13.1%	1.00	0.0%	0.46	0.06	0.0%	—
Nifty 200 Quality 30	12.0%	16.9%	1.24	5.8%	0.37	0.05	-1.1%	0.05
Nifty 200 Momentum 30	15.5%	23.3%	1.57	13.3%	0.46	0.07	1.4%	0.36
Nifty 200 Alfa 30	20.9%	26.3%	1.83	15.4%	0.62	0.09	5.3%	0.67
Nifty 200 Value 30	27.1%	20.9%	1.24	13.5%	0.96	0.16	12.7%	1.05
Gift Re · BiLSTM Top 30	**43.3%**	**34.0%**	**2.31**	**23.2%**	**1.07**	**0.16**	**22.4%**	**1.31**

ASSUMPTIONS

Risk-free rate	*Lag*	*Selection funnel*	*Rebalancing*
6.5%	**60 days**	**1000 → 100 → 30**	**Quarterly**

CAGR VS NIFTY 50

+31.1 pp

43.3% vs 12.2%

RISK-ADJUSTED ALPHA

22.4%

Jensen's α · IR of 1.31

HOW THE MODEL WORKS

1. 13 factor parameters (X1–X13) on Nifty 200 universe
2. Train BiLSTM to predict top-100-of-1000 performers
3. Further Investor Manager shortlist 30 stocks based on his analysis
4. Further weights of the stocks are determined using risk-returns optimization either using Sharpe or Treynor ratios

Solution Proposition 2 — Synthetic CAT Pricing for Rare & Emerging Risks

SP2

When pricing rare or emerging risks, ***synthetic data generated through GANs and Variational Autoencoders*** *will be used alongside real data to understand if it can improve the* ***accuracy, stability, and fairness*** *of risk pricing compared with traditional actuarial and catastrophe-modeling techniques.*

Constructs

- Rare-event frequency & severity
- Tail distribution of losses
- Probable Maximum Loss (PML)
- Pricing stability across cycles

Methods

- GANs, VAEs, WGAN-GP, hybrid models
- EVT-calibrated synthetic scenarios
- Geospatial + demographic + claims data
- Hybrid ML → GLM for explainability

Success metrics

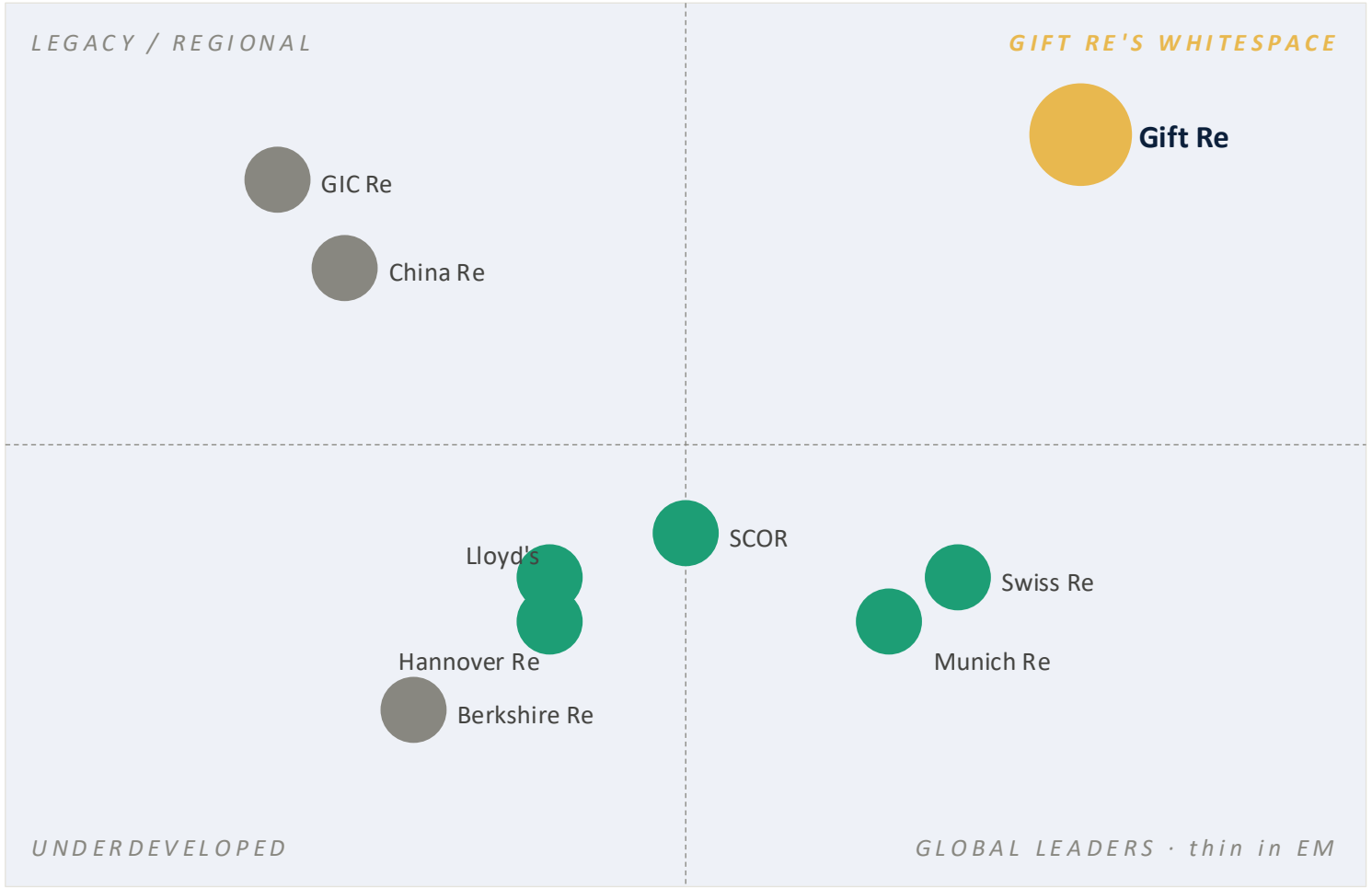
- Tail-loss prediction error vs traditional CAT
- Premium stability across market cycles
- Reduction in adverse-deviation margin
- Protection gap closure (tier-3/4 markets)

PART FOUR

Competition, Financials & Roadmap

Competitive Landscape, Financials, Technology Architecture & MVP Roadmap

Where Gift Re plays — positioning across AI maturity and Emerging Market depth



THE POSITIONING BET

Global leaders have scale and digital maturity but remain thin in tier-3/4 India and wider emerging Asia.

Regional players (GIC Re, China Re) have depth but lag on digitisation and specialty capacity.

Gift Re is purpose-built for the empty top-right quadrant — *AI-native + emerging-market-first.*

Benchmarking AI adoption — Gift Re vs global reinsurers

● High adoption ◑ Partial ○ Low

Score: High=3, Partial=2, Low=1

Reinsurer	Underwriting	Claims	Cat / synthetic	Float / invest.	Platform	Score
Gift Re	●	●	●	●	●	**15/15**
Swiss Re	●	●	◑	○	●	10/15
Munich Re	●	●	◑	○	◑	9/15
SCOR	●	◑	○	○	○	6/15
Hannover Re	●	◑	○	○	○	4/15
Lloyd's	◑	○	◑	○	○	4/15
Berkshire Hathaway Re	○	○	○	○	○	3/15
GIC Re	◑	○	○	○	○	1/15

Incumbents have automated underwriting and claims — none have combined synthetic-data pricing with ML-driven float management. That intersection is Gift Re's white space.

Most global reinsurers have focused AI on underwriting and claims — few have tackled synthetic-data pricing or ML-driven float management. That is Gift Re's white space.

Gift Re's value proposition — four distinct advantages



Synthetic-data pricing

GANs, VAEs & EVT-calibrated tails allow Gift Re to price the catastrophe, climate and cyber risks where competitors either decline or add punitive margins.



ML-driven float

A BiLSTM ensemble trained and tested on 20 years of NSE data delivers alpha over Nifty 200 factor indices — converted directly into ROCE without extra underwriting risk.



AI-ready stack from day one

No legacy mainframes to retrofit. Real-time ALM, automated data pipelines, and containerised ML deployment.



India-first, then other emerging-markets

First-mover with expertise in tier-3/4 India; develop a blueprint which is portable to other emerging economies.

Unit economics — the two engines of a reinsurer

$100	$300	$150	$250	$15	$30
Capital Employed	Premiums Written	Reinsurance Float (~50% of premium)	Total Invested (capital + float)	Investment Income @ ~6% return	Total Return (underwriting + invest)

ENGINE 1 · UNDERWRITING MARGIN

5% of premium @ 95% combined ratio

= $15 on $100 capital → 15% ROCE from underwriting alone. This is Gift Re's H2 lever — synthetic data improves pricing precision and reduces loss ratios.

ENGINE 2 · INVESTMENT INCOME

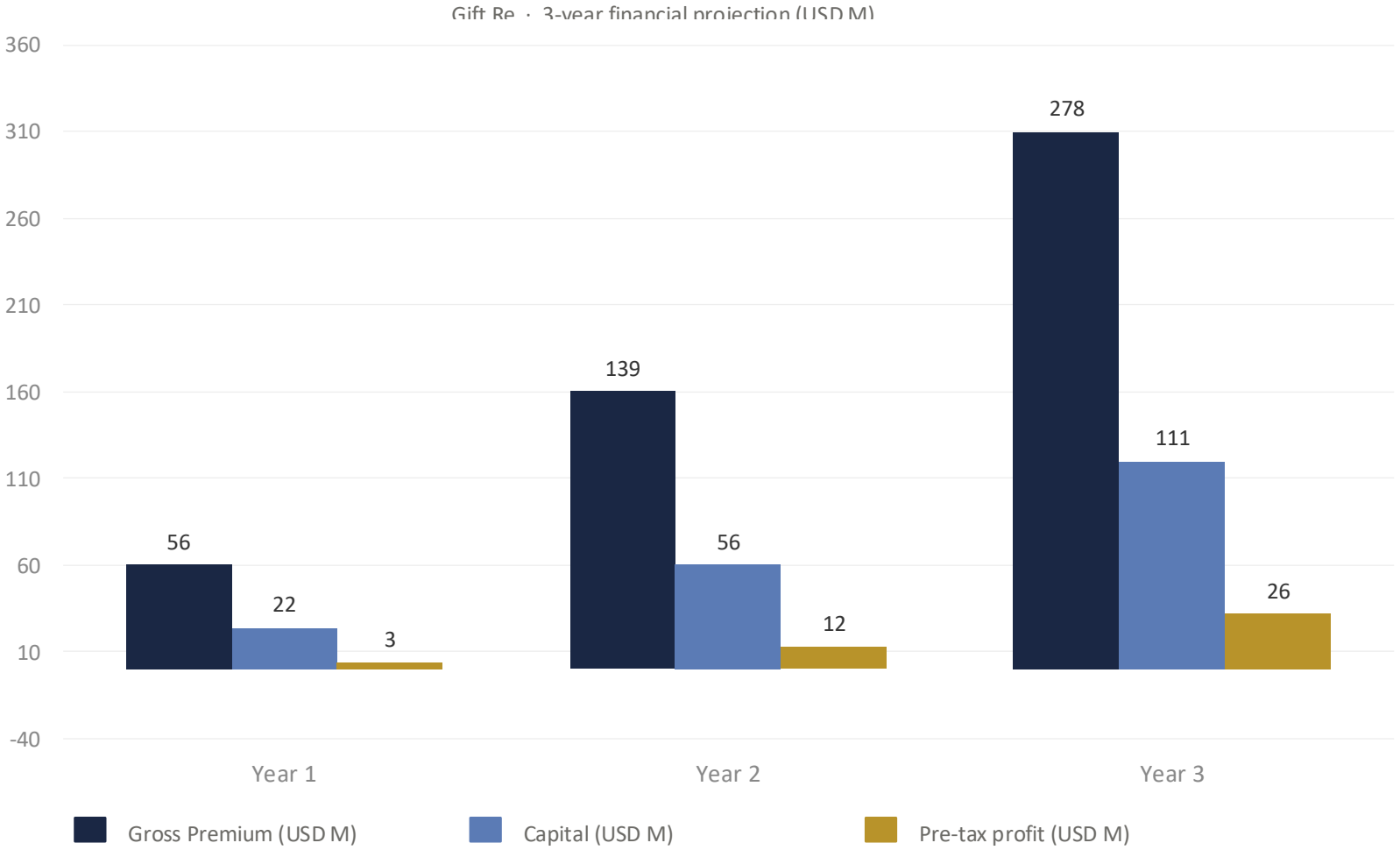
$15 on $250 of assets @ 6% blended yield

= another 15% ROCE on $100 capital. ML based asset allocation and stock selection pushes the yields even higher than 6% higher without breaching solvency or liquidity rules.

Financials of key reinsurers / competitors

	GIC Re	Munich RE	PeakRe
	In ₹ Cr	In ₹ Cr	In USD M
Gross Premium	33,576	6,081	1,556
Commission	6,247	1,037	—
Claims	30,980	4,864	1,367
Investment Income	8,181	506	77
Operational Expense	393	221	58
Profit	3,490	465	208
Capital	40,621	2,060	1,032
Investment Income on Capital	4,611	125	—
Pre-Tax Profit	8,101	589	208
Tax	1,381	58	7.5
Post-Tax Profit	6,497	531	200
ROCE	**16%**	**26%**	**19%**

Financial projections — marginal profits from first year itself, scaling to $300M+ by Year 3



14% → 23% — After-tax ROCE, Y1 to Y3

~40% — Aggregate capital / premium ratio

~50% — Float as % of premium

~95% — Target combined ratio

Financial projections — more details on P&L (₹ Cr)

In ₹ Cr		Year 1	Year 2	Year 3
Gross Premium	2.5× Capital	500	1,250	2,500
Commission	10%	50	125	250
Claims	85%	425	1,063	2,125
Inv. Income – Float	4%	20	50	100
Operational Expense		30	40	50
Profit from Insurance		15	72.5	175
Capital		200	500	1,000
Inv. Income – Capital	8%	16	40	80
Overall Pre-Tax Profit		31	112.5	255
MAT	9%	2.8	10.1	23
Overall Post-Tax Profit		28.2	102.4	232.1
ROCE		**14.1%**	**20.5%**	**23.2%**

Platform architecture — four layers, fully AI-native

01 — OUTPUT LAYER
Management Dashboards · Regulatory Reports · APIs

02 — INVESTMENT ENGINE
BiLSTM · TFT · Random Forest · Portfolio optimisation (Sharpe / Treynor)

03 — RISK ENGINE
GAN / VAE / WGAN-GP · EVT calibration · Cat & cyber scenarios · Pricing APIs

04 — DATA LAYER
Policy & claims · Market data · Satellite / geospatial · Sentiment · IIB · Alternative data

CROSS-CUTTING

Security & governance

ENCRYPTION
at rest and in transit

ACCESS
role-based, least-privilege

AUDIT
every model decision logged

EXPLAINABILITY
hybrid ML → GLM pipelines

COMPLIANCE
IFSCA · IRDAI · GDPR

DEPLOYMENT
containerised, Kubernetes

MVP roadmap — from investment engine to full-stack reinsurer

YEAR 1

MVP 1.0

Foundational Capabilities

FOCUS

Validate the investment engine

- BiLSTM / TFT ensemble trained on NSE top-1,000
- 20 years data on factor indices and their parameters collected and analysed. Back tests to compare the returns of the ML models vis-à-vis the factor indices
- Simple dashboards; internal use only

YEAR 2

MVP 2.0

Enhanced Analytics & Automation

FOCUS

Add synthetic-data pricing

- GAN / VAE pricing models for climate, cyber
- Satellite & geospatial data pipelines
- Pricing + underwriting integration; early client pilots

YEAR 3

MVP 3.0

Full-Scale AI-Driven Reinsurer

FOCUS

Scale and commercialize

- Real-time ALM across asset & liability engines
- API integrations with brokers & cedants
- Expansion beyond India; target $300M+ premium

PART FIVE

Team, Capital & SWOT

Team structure, Risk Based Capital and SWOT analysis

A lean team with cross-disciplinary depth



Risk-Based Capital — the regime Gift Re is purpose-built for

RULE-BASED (CURRENT)	
CAPITAL FORMULA	Flat 150% of required capital, regardless of actual risk
RISK TREATMENT	Same capital charge for low and high-risk books
RARE EVENTS	Handled by conservative margins for adverse deviation
ASSET ALLOCATION	Mostly dictated by regulations and preference for liquid-instruments
MINIMUM REQUIRED	Rs 200 crores to start with but net worth of Rs 1000 crores need to be demonstrated. Same for Gift Re registration.

RISK-BASED (EMERGING)	
CAPITAL FORMULA	Scaled to the actual economic risk of each line
RISK TREATMENT	Capital released on low-risk lines, held on high-risk
RARE EVENTS	Modelled explicitly via stress tests and scenarios
ASSET ALLOCATION	Flexible within solvency and liquidity bounds
MINIMUM REQUIRED	The minimum capital requirements do not change but new regime may lead to efficient allocation → lower capital → higher ROCE

SWOT — what we bring, and where we have to be honest

STRENGTHS

- Deep domain experience across actuarial, investment, technology and sales
- Proven MVP on the investment engine — backtested and benchmarked
- GIFT City regulatory environment purpose-built for reinsurance
- AI-native stack — no legacy systems to migrate

WEAKNESSES

- New entrant — no credit rating at launch
- Limited capital vs global incumbents
- Brand recognition must be earned from zero
- Small team — key-person risk until scale is reached

OPPORTUNITIES

- USD 700B global market growing at 10–11% CAGR
- Tier-3/4 India almost completely underserved by foreign reinsurers
- RBC migration rewards sophisticated modelling
- Cross-border reinsurance from GIFT City opens multiple geographies

THREATS

- GIC Re mandatory cession limits domestic treaty volume
- Regulators may resist synthetic-data-only justifications
- AI talent scarcity and high compensation pressure
- Algorithmic-monoculture risk if many reinsurers adopt similar ML

PART SIX

Business Impact & Key Takeaways

Business Impact & Research Contribution, Risks & Mitigation and Key Takeaways

Business Impact & Research Contribution

ACADEMIC
Theoretical & empirical contributions

- Integrated framework combining EVT + GAN/VAE for reinsurance pricing in a data-scarce emerging market
- Empirical evidence on ML-driven float management — BiLSTM / TFT vs factor-investing benchmarks in Indian equities
- Conceptual model linking asset-side and liability-side AI applications within a single reinsurer balance sheet
- Explainability aspects, where ML is (and is not) acceptable to actuaries and regulators

PRACTITIONER
Operational & strategic contributions

- A blueprint for an AI-enabled reinsurer — architecture, team, economics, and 3-year financial model
- A working MVP and open-source benchmark methodology for backtesting ML portfolios on Indian equities
- A reference implementation of synthetic-data pricing calibrated against EVT tails and useful to regulators
- Documented roadmap for closing the protection gap in tier-3/4 India via alternative-data underwriting

Risks to the research and business venture — with mitigations

RISK	MITIGATION
Regulator resists synthetic-data-only capital relief	Always present synthetic data alongside real experience; use hybrid ML → GLM architectures that actuaries can audit end-to-end.
GAN outputs on genuinely novel risks are unreliable	Restrict synthetic data to rare-but-known perils; flag model confidence explicitly to underwriters and price with conservative loadings for unknowns.
ML portfolio underperforms in a regime shift	Quarterly retraining plus a human investment committee override; backtest across multiple regimes (GFC, COVID) in the dissertation.
Government and Regulators (IRDAI) / GIFT City (IFSCA) support may reduce in future as framework evolves	Gift City is currently gaining prominence with regulatory support, Gift Re will maintain active dialogue with IFSCA team and their working groups to maintain the development initiatives and the momentum.

Key takeaways

1 Underserved market

Global reinsurance is a USD 700B market growing at 10-11%; India at ~15%. Tier-3/4 India is almost unserved by foreign reinsurers.

2 Two solution propositions

SP1: ML-driven float management generates superior risk-adjusted returns on reinsurance capital. SP2: GAN/VAE synthetic data improves pricing of rare catastrophe risks. Both are validated through backtests and expert discovery.

3 Customer discovery & validation

22 senior industry interviews confirmed both pain points. SP1 is backtested with last 20 years data on stocks listed at NSE. SP2 is grounded in EVT theory and expert interview validation — the GAN/VAE pricing pilot remains the next research phase.

4 Good results from the back-tests

BiLSTM portfolio shows alpha over Nifty 200 factor indices — the investment engine is operational.

5 Commercially viable

Financial projections show 14% → 23% after-tax ROCE by Year 3 with $300M+ gross premium.

THANK YOU

Questions, critique, and guidance welcome.

ABHAY TEWARI

Doctor of Business Administration in Emerging Technologies
Golden Gate University · Cohort 6894
Chair: Dr. Siwar Jendoubi
May 2026